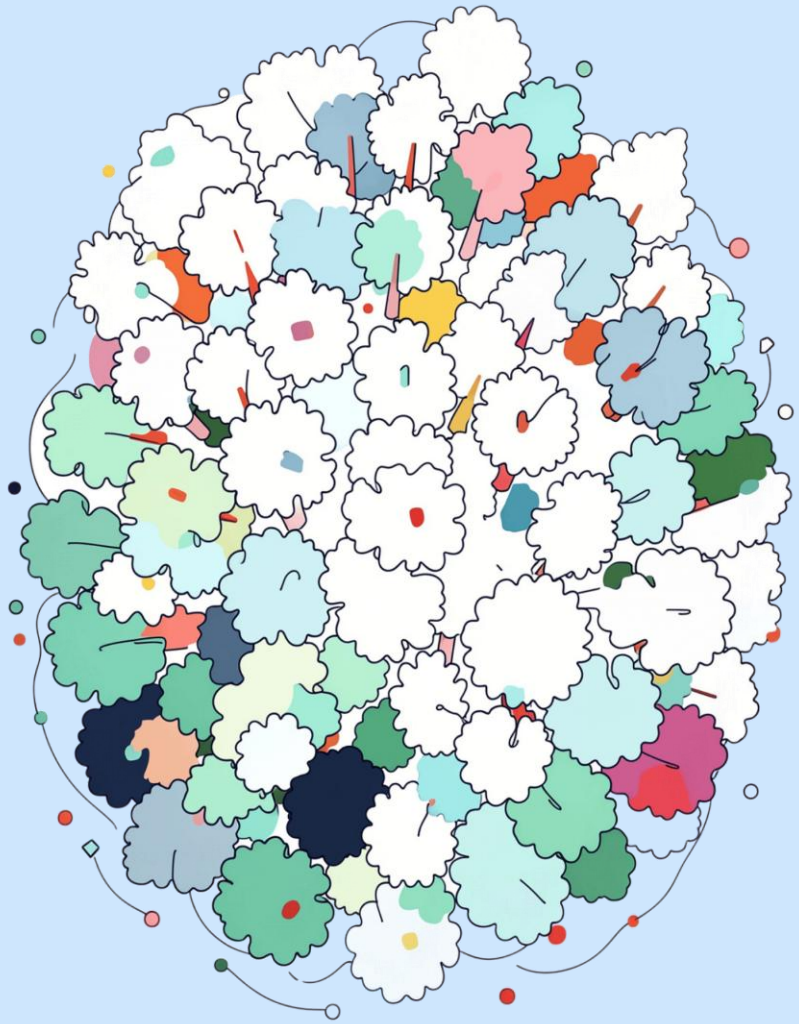
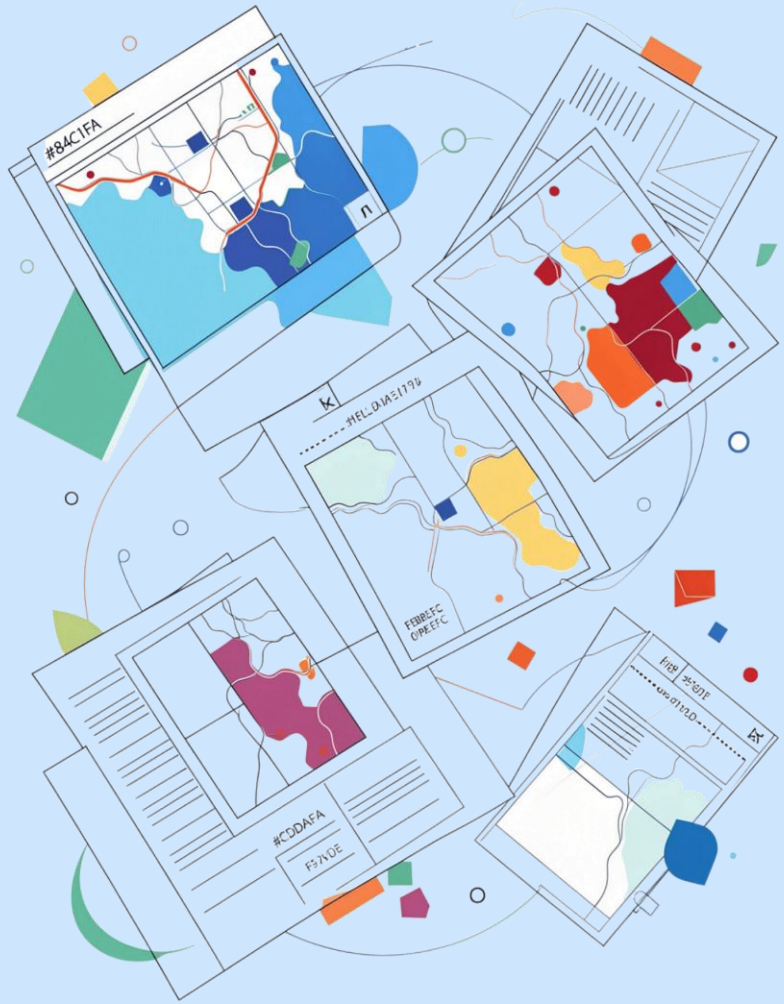


VanRakshak

AI-Powered Forest Intelligence & Response

From Tree-Level Detection to Evidence-Based Forest Decisions

Made with GAMMA



The Problem

Fragmented Workflows

Forest assessment is fragmented across imagery, GIS data, manual field surveys and decision-making

The Key Gap

Connecting tree-level information, geospatial analysis and field verification into one workflow

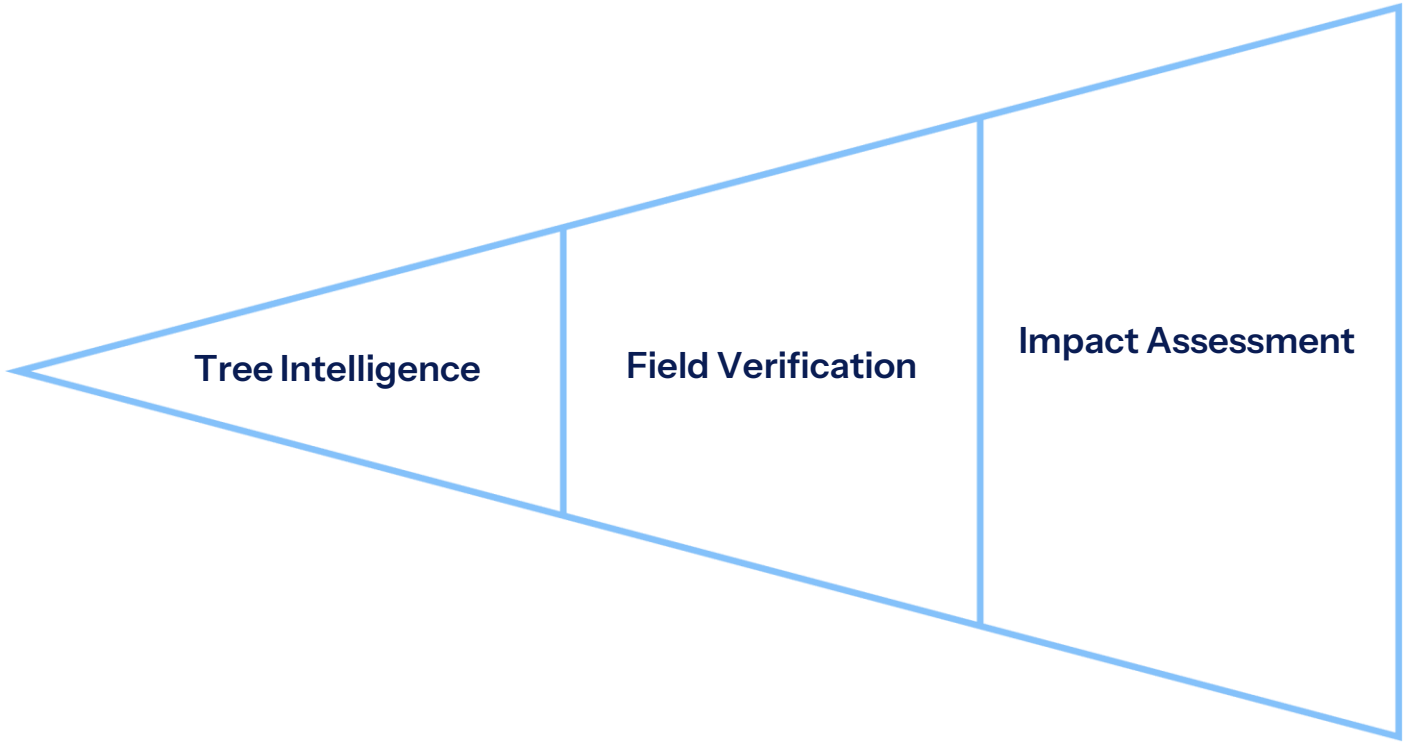
Data exists. The challenge is turning it into verified, actionable intelligence.

Made with GAMMA

The Solution



VanRakshak connects:



It combines AI-powered detection with geospatial analysis and human verification.

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How It Works

Data Input

RGB, LiDAR, GIS and environmental data

AI Detection

YOLOv8 + OpenCV

Tree Intelligence

Individual-tree inventory and analysis

GIS Analysis

Project boundaries, risk and environmental layers

Route Planning

Efficient field-verification route using Dijkstra

Field Verification

GPS + photographs

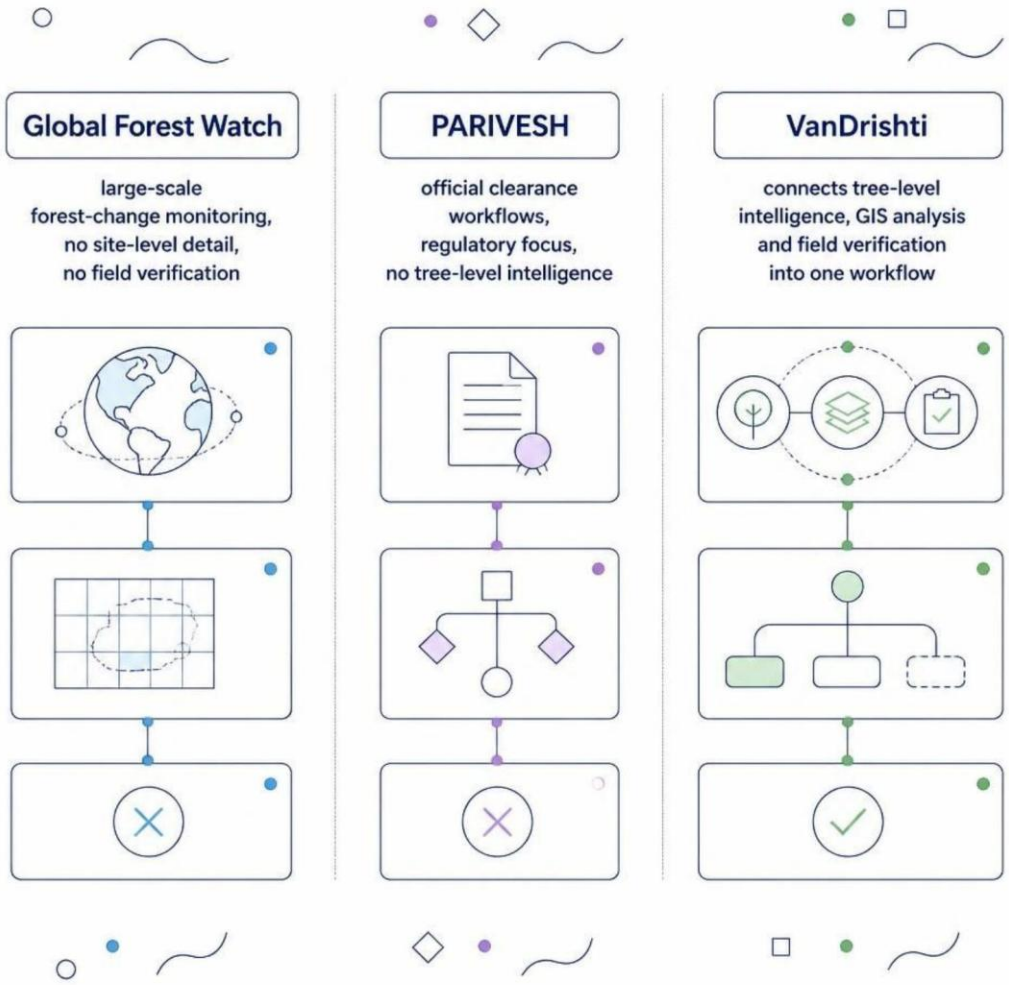
Output

Structured assessment report

Made with GAMMA

What Makes It Different

VanRakshak is not just another forest map. It adds a site-level technical assessment and field-verification layer between data and decisions.



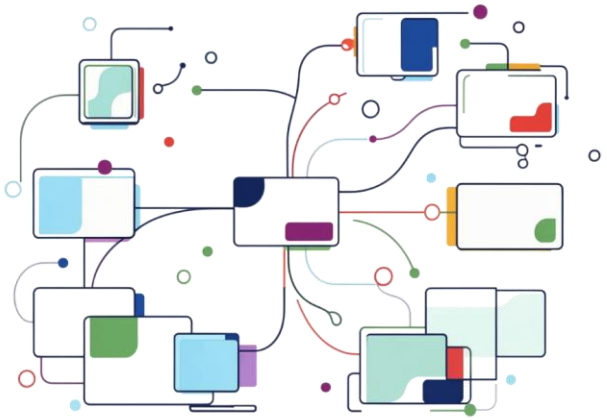
Global Forest Watch
Large-scale forest-change monitoring

PARIVESH
Official clearance workflows

VanDrishti
Connects tree-level intelligence, GIS analysis and verification

Made with GAMMA

Technical Architecture



Frontend

HTML, CSS, JavaScript, Flutter

Backend

Python, Flask

AI & Vision

YOLOv8 · OpenCV

Mapping & Data

Google Maps API · GeoJSON · SQLite

Path Finding

Dijkstra Algorithm

APIs & Reports

NASA FIRMS · OpenWeatherMap · ReportLab / WeasyPrint

Deployment

Cloudflare Worker

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Intelligence Layer

VanDrishti goes beyond static maps through:

Individual Tree Enumeration

Precise per-tree inventory from aerial imagery

Tree Health & Change Analysis

Detect stress, loss and change over time

Species Identification

AI-driven tree species classification

Forest-Fire Information

Real-time fire data via NASA FIRMS

Forest-Risk Mapping

Environmental and boundary risk layers

Optimal Verification Routes

Dijkstra-powered field route planning

STATIC MAPS → TREE-LEVEL INTELLIGENCE → ACTIONABLE WORKFLOW

Made with GAMMA

Human-in-the-Loop



1

AI Identifies

AI identifies and prioritizes observations with confidence scores

2

Humans Verify

Forest officials verify observations using GPS and photographs

3

VanRakshak Documents

Verified information becomes structured assessment data and reports

AI identifies. Humans verify. VanRakshak documents.

Made with GAMMA



Impact

Forest Departments & Field Officers

Faster tree enumeration, site assessment and optimized verification

Researchers & Conservation Agencies

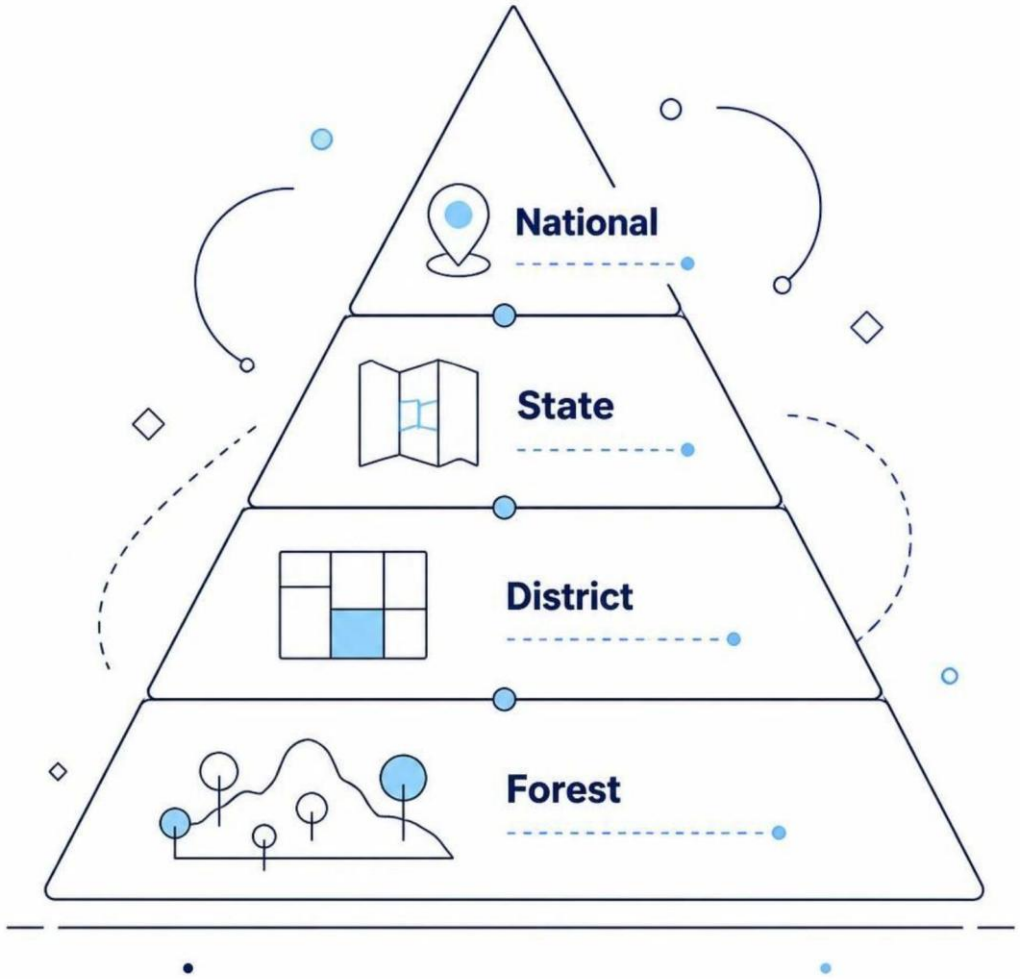
Tree-level datasets for long-term forest monitoring and degradation analysis

Project & Regulatory Authorities

Species-wise, size-wise and health-wise impact information with evidence-backed reports

Made with GAMMA

Feasibility & Scale



1

Technical Feasibility

AI/CV, remote sensing, GIS and modular architecture enable the solution.

2

Challenges & Solutions

Image variability, false positives and data freshness addressed through confidence scoring, multi-modal validation and human verification.

3

Scale Path

Forest → District → State → National

Made with GAMMA

Future Vision

1

Government Workflow Integration

Connect technical assessment outputs with workflows such as PARIVESH

2

Living Digital Forest Twin

Continuously combine imagery, LiDAR and field observations to track individual trees and forest structure over time

3

Broader GIS Intelligence

Extend the platform toward environmental and infrastructure planning

VanDrishti

From Tree-Level Intelligence to Evidence-Based Forest Decisions

Detect → Understand → Verify → Decide

